Sushil Bajpai

Full Stack Software Engineer | MERN · Next.js · LLM/RAG

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Summary

Full Stack Software Engineer with **3+ years** of experience shipping production web and mobile apps with the **MERN stack**, **TypeScript**, **Next.js** and **PostgreSQL**. Built and operate a **multi-tenant SaaS** (payments, realtime, CI/CD) and build **LLM/RAG apps** with hybrid search, reranking and evaluation. Hands-on with **system design** (HLD/LLD) and Agile delivery.

Work Experience

CUBE Technologies **Sep 2025 – Present**

Software Engineer *Gurugram, Haryana, India — Onsite*

- **Owned the Schedules and Tasks modules** end-to-end — Tasks integrates with nearly every other module, demanding careful **cross-module contract design** to ship changes without regressions. Led a **versioning overhaul** of Schedules: a **fault-tolerant version-management system** rolled out safely on live data with **database migrations** and **index/query optimization**.
- Delivered full-stack features across the platform: **React** UI with **state management** and **design-to-code** implementation, plus **role-based authorization**, **API rate limiting**, and **caching** on the backend.
- **Modernized a large legacy codebase**, resolving **long-standing production issues**, simplifying complex workflows, and refactoring fragile components to improve **maintainability, reliability, and delivery velocity**.

EmendoAI, Inc. **May 2025 – Sep 2025**

Software Engineer *Castro Valley, CA, U.S.A. — Remote*

- **Spearheaded** new features for the EmendoAI **React Native** app from **Figma designs** and resolved **100+ critical bugs**.
- **Translated PRDs** into **High-Level and Low-Level Designs (HLD/LLD)**, streamlining feature development.

InterNetwork Expert LLC **May 2024 – May 2025**

Software Engineer *Cary, NC, U.S.A. — Remote*

- **Architected** an interactive CTF (Capture The Flag) challenge platform by integrating **RESTful backend APIs** with responsive frontend interfaces, improving platform usability and interaction by **40%**.
- Designed and implemented **50+** automated test cases using **Python** and **Selenium**, boosting test coverage by **50%**, and deployed isolated virtual lab environments via **Docker** and **Linux** for real-time, hands-on cybersecurity training.

Projects

LocSeva — Restaurant Operating System | *Next.js 15, Fastify, PostgreSQL, Prisma, Redis, Docker* | locseva.satvictech.com

- Built and operate a **multi-tenant restaurant SaaS** end-to-end (**Turborepo**, 45-model **Prisma** schema, **Zod** contracts). Complete guest loop: **book a table → refundable deposit → prepay food → live tracking → automatic close-out billing**.
- Enforced multi-tenant isolation with **Postgres Row-Level Security** and **four scoped JWT token types**; per-tenant **Razorpay/UPI** payment rails (**AES-GCM encrypted**) with **idempotent settlement** and **HMAC-signed** PDF receipts; realtime **KDS/floor dashboards** over **Redis pub/sub**; **GitHub Actions CI/CD** with health-gated auto-rollback.

RAGLab — RAG Pipeline Debugger & Evaluator | *Next.js, TypeScript, Transformers.js, Pinecone, LLMs* | raglabor.vercel.app

- Engineered a **Retrieval-Augmented Generation (RAG)** workbench that visualizes every pipeline stage, with swappable on-device (**Transformers.js**) or cloud providers (**Pinecone**, **OpenAI**, **Claude**, **Gemini**) and 4 chunking strategies.
- Implemented **hybrid search** (BM25 + vector, **Reciprocal Rank Fusion**), cross-encoder **reranking**, tenant filters and **prompt-injection** checks; an **LLM evaluation** harness (10 metrics incl. groundedness, LLM-as-judge) compares configurations.

Portfolio — 10+ Live Projects | *Next.js, React, TypeScript, 3D Web, UI/UX* | sushilbajpai.netlify.app/projects

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, C, C++, SQL
Frontend: React, Next.js, React Native, State Management (Zustand, TanStack Query), HTML/CSS, Tailwind CSS, Figma-to-Code
Backend: Node.js, Express.js, Fastify, NestJS, REST APIs, WebSockets, Authorization (JWT, RBAC), Rate Limiting, Caching
Databases: PostgreSQL (Row-Level Security), MySQL, MongoDB, Redis, Prisma ORM, Schema Design, Indexing, Migrations
Cloud & DevOps: AWS, Docker, GitHub Actions (CI/CD), Vercel, Cloudflare R2, Linux, Git, Postman
Testing & Practices: Playwright, Selenium, API/E2E testing, System Design (HLD/LLD), Agile/Scrum, Code Review, DSA
AI & LLMs: Generative AI, RAG, OpenAI/Claude/Gemini APIs, Prompt Engineering, Embeddings, Vector Databases (Pinecone), Hybrid Search, Reranking, LLM Evaluation, Hugging Face Transformers.js, AI Coding Agents (Claude Code, Codex)

Education

Bharati Vidyapeeth's College Of Engineering, New Delhi **2021 – 2024**

B.Tech in Information Technology & Engineering CGPA: 8.9/10

Achievements

LeetCode: 1100+ Data Structures & Algorithms problems solved, **1000-day solving streak**, 50+ badges leetcode.com/u/sushilbajpai

HackerRank: 5-star rating, skills certifications hackerrank.com/profile/sushilbajpai2003